DDoS / Load-Flood Test Strategy

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> **Reconciled (§11.4.35, 2026-06-26):** the `wg_init_flood` generator (§2, §8) now targets the **canonical plain-WG port `gw:51820`** — a real WG `MessageInitiation` flood hits the plain-WG port that test-rig.md §3.1 drops, not the MASQUE/QUIC `:443` port (the earlier `gw:443` target conflated the two surfaces). The volumetric `iperf3` flood keeps `:443` (the MASQUE/QUIC surface); the edge multiplexes both, but each flood class is modelled on its own port for rig fidelity.

> Master technical specification — Volume 8 (Testing & QA), nano-detail document **ddos**, one of the seven §11.4.169 cross-cutting test-type deep-dives. It deepens `10-testing-acceptance-and-qa.md` §5.8 (DDOS) and the §2 taxonomy row marked `NOT_APPLICABLE: single-node-selfhost` at MVP into an implementation-ready, **author-now-parked** (decision QA-D2, overview §10) bank that re-arms mechanically when Phase 2's public multi-tenant surface exists. SPEC-ONLY: it describes the harness, the fixtures, the captured evidence, the acceptance gate, and the paired §1.1 mutation; it does not build the product. Sources cited inline by id — `[OVERVIEW]` = doc 10; `[TM]` = `[../v05-security/threat-model.md]`; `[01-DP]` = doc 01 data plane; `[04_P1]`/`[04_P2]` = the Phase-1/Phase-2 refined docs; `[svc-policy]`/`[svc-pki]` = the Volume-3 nano-docs. Any claim not grounded in the evidence base is marked **UNVERIFIED** per constitution §11.4.6 — never fabricated.

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0. Scope on HelixVPN surfaces + MVP status

A VPN gateway is, by definition, an Internet-reachable target: the **Rust edge** listens on `:443` for WG/MASQUE datagrams ([TM §5.3], `T-EDGE-D-1`) and the **Go control plane** answers enroll / `WatchNetworkMap` / policy RPCs over mTLS ([TM §5.2]). DDoS is the threat that *no single client* and *no flood of init packets* may exhaust those surfaces. At MVP the deployment is a single rootless-Podman host with no public multi-tenant surface [04_P1 §11]; a volumetric DDoS at MVP would exercise a threat the topology does not yet present, so the coverage-ledger cell is `NOT_APPLICABLE: single-node-selfhost` (§11.4.6, overview §6.3 row F-DDOS-FLOOD) — an **honest, reasoned** NA, never a silent omission.

The seam, however, is built **now** so the suite re-arms by data, not by code: the per-tenant / per-source rate-limit token buckets in Redis ([TM §5.1 `T-CONN-D-1`], [svc-pki §3.4]), WG's stateless cookie-reply anti-DoS ([01-DP], [TM §5.3 `T-EDGE-D-1`]), and the stateless fail-static edge that drops and rootless-auto-restarts rather than crashing [01-DP I3]. This document authors the bank against those seams; QA-D2 records the decision (author-now-parked).

Surface	Flood class	MVP seam (built)	Phase-2 re-arm
edge `:443` data port	WG/MASQUE handshake-init flood	WG cookie-reply; host firewall floor	full handshake-flood bank
edge `:443` data port	volumetric byte flood	stateless drop + rootless restart [01-DP I3]	upstream-scrubbing-assisted bank
control plane mTLS RPC	enroll / `SignDeviceCert` flood	Redis token bucket, per-tenant + per-source [svc-pki §3.4]	KMS-quota-protection bank
control plane	`WatchNetworkMap` stream-open flood	bounded streams [svc-policy §11]	coordinator fan-out soak
control plane	policy compile-bomb	bounded `O(devices×rules×targets)` [TM §5.2 `T-CP-D-1`]	10k×1k tenant soak

1. The two threats this suite proves bounded

The bank is built around two falsifiable claims, each phrased so the PASS is a captured-evidence proof that a **legitimate** client is unharmed *while the flood runs* (anti-bluff: a flood test that only proves "the flood happened" is a §11.4 bluff — it must prove the service stayed usable).

- **D-FLOOD-HANDSHAKE.** A torrent of WG/MASQUE init packets (and a torrent of unauthenticated enroll attempts) must not exhaust the gateway.

Claim: during the flood, a *legitimate* enrolled client's handshake p99 latency stays under its SLO budget, the edge process never OOM/crashes (`helix_edge_up == 1` throughout), and `wg_init_dropped` / `ratelimit_rejected` counters rise (proving the rate limiter *engaged*, not that the flood was absent).

- **D-FLOOD-VOLUMETRIC.** A byte flood on `:443` must make the stateless edge **fail-static** — drop excess, never crash, rootless-auto-restart if the supervisor cycles it [01-DP I3]. **Claim:** the edge liveness counter stays 1 (or restarts within the supervisor budget with zero plaintext leak during the gap — composes the kill-switch S9 evidence, [TM §5.4 `T-CLI-T-1`]).

Each claim is a `challenges` bank entry (overview §5.5) scored on captured evidence, never the generator's exit code.

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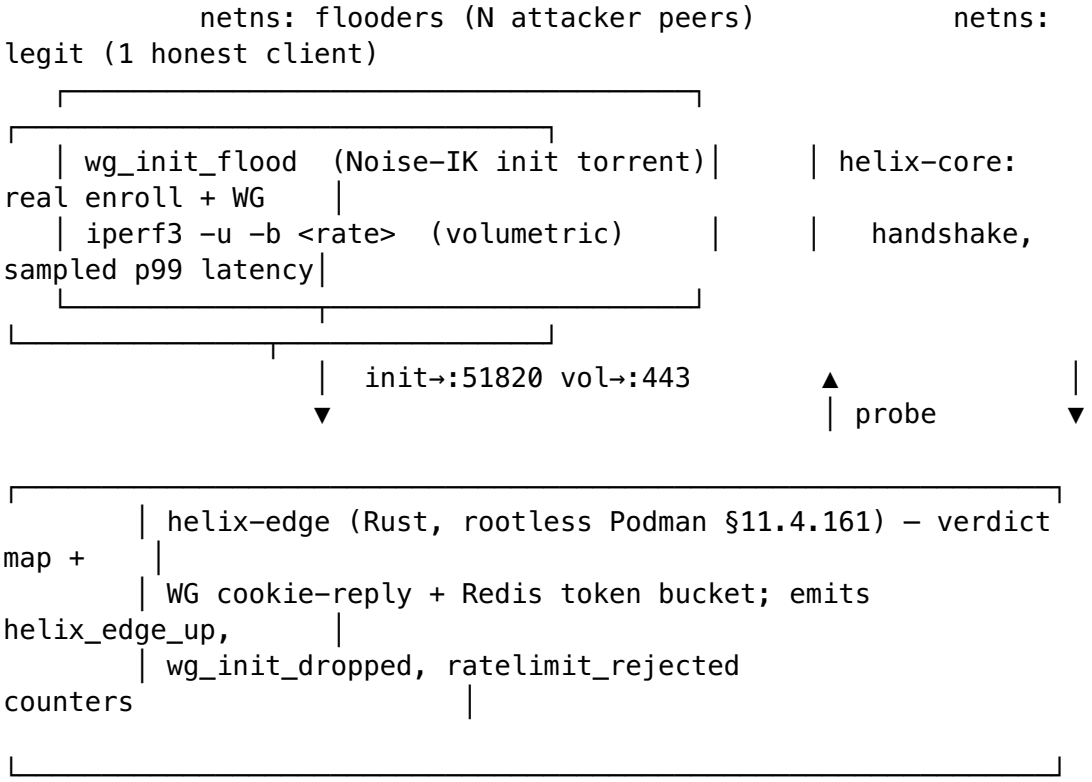
2. Harness — the flood rig

The rig extends the netns rig (overview §5.3, [`test-rig.md`]) with two flood generators and a **legitimate-client probe** that runs *concurrently with* the flood — the probe is the thing the acceptance gate measures. Single-resource-owner partitioning (§11.4.119): exactly one process owns the edge data port; the flood generators and the legit probe are distinct netns peers.



- **`wg_init_flood`** — a custom Noise-IK initiation-message generator (not a mock: it emits real WG `MessageInitiation` frames with valid framing but unregistered static keys, exactly the cheapest attacker packet [01-DP], [TM `T-EDGE-D-1`]); it targets the **canonical plain-WG port `udp/51820`** (the port [`test-rig.md` §3.1] drops), *not* the MASQUE/QUIC `:443` surface — plain WG is `udp/51820`, MASQUE/QUIC is `:443` (the edge multiplexes both, but the

handshake-flood models real plain WG on its own port). Rust binary `rig/wg_init_flood`, rate configurable (`--pps`).
- **`iperf3 -u -b <rate>`** — UDP volumetric flood at line rate toward the MASQUE/QUIC `:443`.
- **legit probe** — a real `helix-core` instance performing genuine enroll + WG handshake + `iperf3` goodput in the `legit` netns, sampling handshake completion latency.
- **counter scrape** — `helix-edge`'s Prometheus endpoint (control-plane-local, no per-user data, invariant I5) is scraped at 1 s cadence into a CSV.

The rig runs under `containers/pkg/boot` for Redis (the token-bucket backing store) — the sole container path (§11.4.76); the netns peers need `CAP_NET_ADMIN` (the documented scoped sudo exception, overview §9), never container-management escalation.

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3. Fixtures — real, no mocks (§11.4.27)

DDoS is an integration/e2e-class type: **no mocks** below the unit layer. Fixtures are real:

Fixture	What it is	Why real
`legit_client.toml`	a genuinely enrolled `helix-core` config (real device cert, real WG key)	the probe must complete a *real* handshake — a mocked client cannot prove the limiter spares legitimate traffic
`attacker_keys.bin`	N=10k unregistered Curve25519 static keys for `wg_init_flood`	real WG frames with keys the edge will reject — the cheapest real attacker packet
`tenant_ratelimit.yaml`	the real Redis token-bucket config under test [svc-pki §3.4]	the limiter's real thresholds, not a stubbed allow-all
`edge_image`	the published rootless-Podman edge image (S8)	the suite must flood the *shipping* artifact (§11.4.108), not a debug build

The legitimate-client fixture is the load-bearing one: the suite's entire value is proving the **real** honest client is unharmed; a stubbed probe would be a B3 wrong-plane bluff.

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4. Captured evidence (§11.4.69 / .85)

Every PASS cites artifacts under `qa-results/ddos/<run-id>/`, scored by the `challenges` engine, never the generator's exit code. The §11.4.69 evidence class is `network_throughput` / `network_connectivity` for the legit probe and a liveness counter for the edge.

Artifact	Shape	Asserts
`legit_latency.json`	per-sample handshake-complete + goodput series, p50/p95/p99	legit p99 ≤ budget *during* the flood
`edge_liveness.csv`	1 Hz `helix_edge_up`, RSS, FD count	edge never 0 (or restart within supervisor budget)
`limiter_counters.csv`	`wg_init_dropped`, `ratelimit_rejected` time series	the limiter *engaged* (counters rose) — proves the flood reached the edge, not that it was absent
`gap_pcap.pcap`	host-iface capture during any edge restart	zero plaintext / zero `:53` during the gap (composes S9, [TM `T-CLI-T-1`])
`categorised_errors.txt`	per-error-class counts from the flood window (§11.4.85)	refused-cleanly vs crashed classification

Anti-bluff rule (overview §0, B2): "no error during flood" is **not** evidence — the limiter counters MUST show engagement and the legit-probe latency CSV MUST show bounded service, or the test is a B2 absence-of-error bluff and FAILs.

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5. Determinism (§11.4.50)

The flood is inherently noisy, so determinism is asserted on the **verdict**, not the raw rate. `ab_run_n_times "ddos-handshake" 3 run_handshake_flood` runs the bank N=3 against the same edge image MD5 + same rig; the per-run evidence-hash is computed over the *verdict tuple* `(edge_stayed_up: bool, legit_p99_within_budget: bool, limiter_engaged: bool, leak_during_gap: bool)` — all three runs MUST yield the identical verdict tuple. A run where the edge stays up in 2 of 3 is **auto-FAIL** (§11.4.50: no first-pass-was-a-flake escape). Cycle-validation runs N=10.

The raw rate/latency numbers are recorded for trend (§11.4.24 min/max/mean/p95) but are not the determinism key — the *verdict* is, because the property under test is binary ("bounded / not").

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6. Acceptance gate

#	Bar	Evidence	Phase
DDOS-G1	legit handshake p99 ≤ SLO budget while flood at ≥ design-attack-rate runs	`legit_latency.json`	Phase 2

#	Bar	Evidence	Phase
DDOS-G2	`helix_edge_up == 1` throughout, OR restart ≤ supervisor budget with **zero** leak in `gap_pcap`	`edge_liveness.csv` + `gap_pcap.pcap`	Phase 2
DDOS-G3	limiter engaged: `ratelimit_rejected`/`wg_init_dropped` strictly > 0 during flood	`limiter_counters.csv`	Phase 2
DDOS-G4	control-plane enroll-flood: legit enroll still succeeds; KMS `SignDeviceCert` quota not exhausted	enroll-success log + quota counter	Phase 2

At MVP the gate is recorded `NOT_APPLICABLE: single-node-selfhost` in the coverage ledger with the §11.4.3 reason; the gate is **release-blocking in Phase 2** when the multi-tenant surface exists. A gate that cannot clear in its time box **is the finding** — escalate per §11.4.66, never overrun silently.

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7. The paired §1.1 mutation

The mutation proves the gate is not a bluff — it must FAIL the gate when the protection is removed:

```
MUTATION (paired §1.1, gate CM-DDOS-RATELIMIT-ENGAGED):
  Disable the Redis token-bucket check in the edge enroll/
handshake path
  (the `ratelimit.Allow()` call returns true unconditionally).
EXPECTED:  under the flood, the legit probe's p99 blows past
budget AND
           limiter_counters.csv shows ZERO ratelimit_rejected
           → DDOS-G1 + DDOS-G3 FAIL → mutation caught.
RESTORE:   re-enable ratelimit.Allow(); re-run → GREEN.
```

A second mutation (`CM-DDOS-EDGE-FAILSTATIC`) replaces the stateless drop with an unbounded allocation per init packet; expected: edge RSS climbs to OOM under the flood, `helix_edge_up` drops with no clean restart → DDOS-G2 FAILs → caught. Both mutations are restored and the working tree is verified quiescent (§11.4.84) before any commit.

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8. Test skeletons

```rust
// rig/wg_init_flood/src/main.rs — REAL Noise-IK init torrent (not
        a mock), §11.4.27
fn main() -> anyhow::Result<()> {
    let args = Args::parse();                  // --target :443  --
        pps  --keys attacker_keys.bin
```

```
    let keys = load_static_keys(&args.keys)?; // 10k unregistered
        Curve25519 keys
    let sock = UdpSocket::bind("0.0.0.0:0")?;
    let pacer = Pacer::new(args.pps);
    loop {
        let k = keys.next_round_robin();
        let init = wg::MessageInitiation::new(&k,
        &args.gateway_pubkey); // valid framing, bad identity
        sock.send_to(&init.encode(),
        args.target)?;                          // the cheapest
        attacker packet
        pacer.tick();                             // bounded send
        rate; never floods the *host* (§12 host-safety)
    }
}

# rig/ddos_handshake.sh — the D-FLOOD-HANDSHAKE challenge driver
set -euo pipefail
out="qa-results/ddos/$(date +%s)"; mkdir -p "$out"
trap 'rig/flood_stop.sh; rig/netns_down.sh' EXIT            #
        §11.4.14 cleanup on every path
scrape_counters_1hz "$out/limiter_counters.csv" &
        # edge counters
sample_edge_liveness_1hz "$out/edge_liveness.csv" &         #
        helix_edge_up, RSS
ip netns exec flooders rig/wg_init_flood --target gw:51820 --pps
        "$ATTACK_PPS" --keys attacker_keys.bin &  # plain-WG port
        (test-rig.md §3.1)
sleep 2
        # let the flood ramp
ip netns exec legit helix-core probe --enroll legit_client.toml --
        iperf 20 \
    --latency-out "$out/legit_latency.json"                 # the
        LEGIT client, concurrently
p99=$(jq '.handshake_p99_ms' "$out/legit_latency.json")
up=$(min_col "$out/edge_liveness.csv" helix_edge_up)        # min
        over the window
rej=$(max_col "$out/limiter_counters.csv" ratelimit_rejected)
[ "$up" = 1 ] && within_budget "$p99" && [ "$rej" -gt 0 ] \
  && ab_pass_with_evidence "edge bounded under handshake flood;
        legit p99=$p99ms" "$out" \
  || ab_fail "DDoS bound breached: up=$up p99=$p99 rejected=$rej"

# challenges/banks/helixvpn_phase2_ddos.yaml — parked, re-arms in
        Phase 2 (QA-D2)
bank: helixvpn-phase2-ddos
arm_condition: deployment.topology == "multi-tenant-ha"   #
        mechanical re-arm, not a code edit
challenges:
  - id: HVPN-CHAL-DDOS-HANDSHAKE
    feature_class: network_connectivity
    driver: rig/ddos_handshake.sh
    evidence:
      latency: { path: "legit_latency.json", assert:
        "handshake_p99_ms <= slo.handshake" }
```

```
    liveness:{ path: "edge_liveness.csv",  assert_min:
      "helix_edge_up == 1" }
    limiter: { path: "limiter_counters.csv", assert:
      "max(ratelimit_rejected) > 0" }   # defeats B2
  self_validated: true                                  #
    golden-bad: limiter-off run MUST score FAIL
```

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9. The MVP SKIP record + re-arm trigger

At MVP the coverage ledger carries (overview §6):

```
feature_id = F-DDOS-FLOOD
  DDOS cell: state = NOT_APPLICABLE, skip_reason = 'single-node-
selfhost'  -- §11.4.3 closed reason
  CHAL cell: state = NOT_APPLICABLE, skip_reason = 'single-node-
selfhost'
```

The re-arm is **mechanical** (overview §6.2 state machine, §7.3): when the deployment topology changes to `multi-tenant-ha` in Phase 2, the `arm_condition` flips the cell from `NOT_APPLICABLE` to `REQUIRED`, the parked bank activates, and the gate becomes release-blocking. No code change is needed to re-arm — the bank already exists (QA-D2), satisfying the §11.4.6 "the seam exists now" discipline so Phase-2 cannot ship a DDoS gap by omission.

UNVERIFIED. The concrete design-attack-rate (`ATTACK_PPS`), the legit-handshake SLO budget under flood, and the supervisor restart budget are Phase-2 measured numbers, not yet results; they are stated as Phase-2 targets here and become facts only when the Phase-2 rig produces the CSVs. The WG cookie-reply effectiveness against the specific init-flood rate is asserted from the WireGuard design [01-DP] but **UNVERIFIED** for HelixVPN's edge until the Phase-2 bank runs.

Sources verified

- [OVERVIEW] [`../10-testing-acceptance-and-qa.md`] — §2 taxonomy (DDOS `NOT_APPLICABLE` row), §5.8 (DDoS strategy), §6.3 (F-DDOS-FLOOD ledger row), §7.3 (Phase-2 re-arm), §10 QA-D2. (Read 2026-06-26.)
- [TM] [`../v05-security/threat-model.md`] — `T-EDGE-D-1` (handshake/UDP flood), `T-CONN-D-1`, `T-CP-D-1` (compile-bomb), `T-PKI-D-1` (KMS quota), §10 R-DOS residual. (Read 2026-06-26.)
- [01-DP] `final/01-data-plane.md` — WG stateless cookie-reply anti-DoS, stateless fail-static edge (I3); cited via the overview + threat-model.
- [svc-pki] [`../v03-control-plane/svc-pki.md`] / [svc-policy] [`../v03-control-plane/svc-policy.md`] — §3.4 enroll rate limits, §11 bounded compile/streams; cited via the threat-model.
- [04_P1 §11] single-node MVP topology (the basis for the MVP `NOT_APPLICABLE`); [04_P2] Phase-2 HA surface — cited via the overview.
- Constitution: §11.4.169 (test-type mandate), §11.4.3 (honest SKIP/NA reason), §11.4.6 (no-guessing / UNVERIFIED), §11.4.27 (no-fakes), §11.4.69 (sink-side evidence), §11.4.85 (stress/chaos taxonomy adjacency), §11.4.50 (determinism),

§11.4.84 (quiescence), §11.4.119 (single-resource-owner), §11.4.108 (flood the shipping artifact), §1.1 (paired mutation).